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BIAS VOLTAGE ADJUSTMENT FOR A PHASE LIGHT MODULATOR

Abstract

An integrated circuit includes an electrode voltage controller, a micro-electromechanical system (MEMS) structure, and a bias voltage generator. The MEMS structure has a first electrode, a conductive plate, and a reflective layer on the conductive plate. The first electrode is coupled to the electrode voltage controller, and the conductive plate is configured to move vertically with respect to the first electrode responsive to a voltage generated by the electrode voltage controller and applied to the first electrode. The bias voltage generator is coupled to the conductive plate. The bias voltage generator has an input configured to receive a bias control signal. The bias voltage generator is configured to apply a non-zero bias voltage to the conductive plate responsive to the bias control signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 17/120,687 filed Dec. 14, 2020, which claims priority to U.S. Provisional Application No. 62/947,217, filed Dec. 12, 2019, which is hereby incorporated by reference.

BACKGROUND

[0002] Actuators (e.g., electrostatic and non-electrostatic) are used in various technologies. For example, actuators may be used in phase light modulators (PLMs) to modulate the phase of light. Phase light modulators can be implemented as microelectromechanical systems (MEMS) that include an array of mirrors. Incident light beams reflect off the mirrors. The MEMS mirrors can be independently, vertically moved to vary the phase of the incident light beam. Each mirror may represent a pixel. Each pixel in such microelectromechanical actuator systems includes a base electrode as well as a spring electrode coupled to the mirror. When a voltage differential is created between the base electrode and the spring electrode, the spring electrode moves towards the base electrode, thereby moving the mirror to a different position. Such microelectromechanical system phase light modulators are used in a variety of applications such as high dynamic range cinema, light detection and ranging systems, high volume optical switching (e.g., used in telecom or server farms), microscopy/spectroscopy/adaptive optics (e.g., used in astronomy, ophthalmology, machine vision, etc.), and holographic displays.

SUMMARY

[0003] In one example, an integrated circuit includes an electrode voltage controller, a micro-electromechanical system (MEMS) structure, and a bias voltage generator. The MEMS structure has a first electrode, a conductive plate, and a reflective layer on the conductive plate. The first electrode is coupled to the electrode voltage controller, and the conductive plate is configured to move vertically with respect to the first electrode responsive to a voltage generated by the electrode voltage controller and applied to the first electrode. The bias voltage generator is coupled to the conductive plate. The bias voltage generator has an input configured to receive a bias control signal. The bias voltage generator is configured to apply a non-zero bias voltage to the conductive plate responsive to the bias control signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 shows an example of a phase light modulator having a mirror plate that moves vertically with respect to an electrode as a function of an applied voltage to the electrode.

[0005] FIG. 2 shows an example implementation of the electrode.

[0006] FIGS. 3 and 4 illustrate two different vertical positions of a mirror within a phase light modulator.

[0007] FIG. 5 shows an example of a bias voltage generator usable in the phase light modulator.

[0008] FIGS. 6-8 illustrate an example in which the bias voltage generator adjusts the bias voltage of the PLM for wavelength control.

[0009] FIG. 9 illustrates an example in which the bias voltage generator adjusts the bias voltage of

the PLM for temperature compensation.

[0010] FIGS. **10-12** illustrate an example in which the bias voltage generator adjusts the bias voltage of the PLM for speckling control.

[0011] FIG. **13** illustrates an example in which the bias voltage generator adjusts the bias voltage of the PLM for laser drift compensation.

[0012] FIG. **14** illustrates an example in which the bias voltage generator adjusts the bias voltage of the PLM for vibration compensation.

[0013] FIGS. **15-16** illustrate an example in which the bias voltage generator adjusts the bias voltage of the PLM for binning control.

[0014] FIGS. **17A** and **17B** illustrates an example in which the bias voltage generator adjusts the bias voltage of the PLM for modulation over wide bandwidth illumination sources.

[0015] FIGS. **18A-18C** illustrates an example in which the bias voltage generator adjusts the bias voltage of the PLM for charge mitigation.

[0016] FIG. **19** illustrates an example in which the bias voltage generator adjusts the bias voltage of the PLM for conjugate ghost image control.

[0017] FIG. **20** illustrates the generation of a bias voltage based on multiple factors.

DETAILED DESCRIPTION

[0018] As described above, to modulate a beam of light, a phase light modulator (PLM) includes mirrors that can be adjusted (e.g., moved or displaced) in order to change the properties (e.g., phase) of a reflected beam of light. In some examples, spatial light modulators use actuators to move the mirrors responsive to applied voltages. In some examples, PLMs use microelectromechanical systems (MEMS)-based actuators to move the mirrors based on a combination of an electrostatic force and a spring force.

[0019] A parallel-plate, electrostatic actuator (e.g., used in a MEMS) is a device that utilizes electrostatic force to move an object (e.g., a mirror of a phase light modulator pixel). For example, the actuator includes a movable conductive plate that supports a mirror. The conductive plate is also anchored to an attachment at a distance d from a fixed electrode. The conductive plate includes one or more flexural arms attached to support posts. The flexural arms function as a spring (having a spring constant, k) and contribute to a spring constant, k (stiffness). The conductive plate and the electrode are parallel to each other, and a potential difference is applied between the electrode to force them nearer together or farther apart, hence the name “parallel-plate actuator.”

[0020] Responsive to an applied potential difference (which creates an electrostatic force) between the electrode and the conductive plate relative to the spring implemented by the conductive plate and its flexural arms, the conductive plate moves towards (or away from) the electrode. Usually, the conductive plate is coupled to ground, and the electrode is coupled to a voltage regulator. The voltage regulator applies a variable voltage to the electrode. When the voltage applied to the electrode increases, the voltage differential between the electrode and the conductive plate generates an electrostatic force that drives the conductive plate towards the electrode, thereby moving the mirror (which is supported on the conductive plate) toward the second electrode.

[0021] Alternatively, the electrode may be implemented as a digital electrode having multiple conductive portions, each capable of separately receiving a particular voltage. Accordingly, the combined surface area of the conductive portions subject to an applied voltage can vary. As the amount of area of the digital electrode receiving the applied voltage increases, the electrostatic force also increases resulting in the conductive plate being pulled nearer digital electrode. As the voltage decreases (and/or the amount of area applying the voltage on the digital electrode decreases), the electrostatic force decreases resulting in the conductive plate moving away from the digital electrode (e.g., due to the mechanical force of the flexural arms). In this manner, a controller can control the voltage and/or amount of area receiving a voltage on the digital electrode to control the position of the conductive plate, thereby controlling the position of the mirror. The amount of travel of a mirror corresponds to an achievable phase modulation of the PLM.

[0022] The term “electrode” as used herein may refer to a single conductive element that can receive a variable voltage. The term “electrode” also may refer to a digital electrode having multiple conductive area portions, each of which individually receives a particular voltage.

[0023] FIG. 1 shows a three-dimensional view of a MEMS structure **100** for a single element of a phase light modulator. An integrated circuit may have an array of such MEMS structures **100** to form the phase light modulator. As shown in the example of FIG. 1, the MEMS structure **100** includes electrode **110**, conductive plate **106**, a mirror **102**, and support posts **108**. The support posts **108** are electrically conductive (e.g., metal) and extend upward from a metal structure **109** (sometimes referred to as the bottom metal). MEMS structure **100** includes four support posts **108**. The conductive plate **106** is generally square but can have any suitable shape. The conductive plate **106** includes one or more (four in the example of FIG. 1) flexural arms **107**. The flexural arms **107** of the conductive plate **106** have a mechanical spring constant. When stretched, the flexural arms **107** apply a mechanical force in the opposite direction of the stretching. The flexural arms **107** are attached to the support posts **108**. The support posts **108**, flexural arms **107**, and conductive plate **106** are all formed of an electrically conductive material (e.g., a metal) and coupled together (or formed as a unitary set of components). Accordingly, the conductive plate **106** will be at the same potential as the support posts **108**.

[0024] The mirror **102** is a reflective layer and is supported above the conductive plate **106** by one or more mirror attachments **104**. Each mirror attachment **104** may be a via formed within the MEMS structure. A support plate (not shown in FIG. 1) may be disposed between the mirror attachments **104** and the mirror **102**. Accordingly, the mirror **102** is supported by the support plate, and the support plate is supported by the mirror attachments **104**. In some examples, a different object may be included in place of the mirror **102**.

[0025] FIG. 2 shows an example of electrode **110** within metal structure **109**. The electrode **110** and the metal structure **109** may be formed from the same metal layer which is etched to the form the structures shown in FIG. 2. The electrode **110** in this example is a digital electrode including an inner electrode portion (E1) **201**, a middle electrode portion (E2) **202**, and an outer electrode portion (E3) **203**. While three electrodes forming the digital electrode **110** are shown in this implementation, any suitable number of electrodes may be included.

[0026] FIGS. 3 and 4 illustrate the mirror at two different vertical positions relative to the conductive plate **106**. An electrode voltage controller **112** is shown coupled to electrode **110**. FIG. 3 illustrates the vertical position of the conductive plate **106** (and thus mirror **102**) with the voltage on electrode **110** from the electrode voltage controller **112** at a voltage level that is smaller than the voltage on electrode **110** in FIG. 4. Because the electrode voltage is larger in FIG. 4 than in FIG. 3, the conductive plate **106** is nearer the electrode **110** in FIG. 4 than in FIG. 3.

[0027] Although FIGS. 3 and 4 illustrate two positions for the conductive plate **106** relative to the electrode **110**, the potential difference applied between the conductive plate **106** and the electrode **110** can be varied to cause the conductive plate **106** to be at any of multiple (two, three, four, or more) different positions relative to the electrode **110**. In the example of FIG. 2, with electrode **110** including three electrode portions **201-203** and each electrode portion receiving either ground or a fixed voltage, eight different electrode configurations are possible and thus eight different separation distances between the conductive plate **106** and the electrode **110** are possible.

[0028] As described above, for some phase light modulators, the support posts **108** is connected to electrical ground. Consequently, the conductive plate **106** also is grounded. However, various benefits can be realized if the conductive plate receives a bias voltage other than ground. To that end, FIGS. 3 and 4 illustrate that the integrated circuit containing the MEMS structure **100** also includes conductive plate bias voltage generator **120** (also referred to as a bias voltage generator). The bias voltage generator **120** includes an input **121** which can receive a bias control signal **220**. The bias voltage generator **120** generates a bias voltage **130** based on information encoded in the bias control signal **220**. In one example, the bias voltage **130** is a direct current (DC) voltage

having a magnitude based on the bias control signal **220**. In other examples, the bias voltage **130** is a time varying voltage having a magnitude and/or frequency based on the bias control signal **220**. [0029] FIG. 5 shows an example implementation of the bias voltage generator **120**. In this example, the bias voltage generator **120** includes a bias voltage determination circuit **510** coupled to a voltage regulator **520**. The bias voltage determination circuit **510** includes the input **121** that receives the bias control signal **220**. An output **511** of the bias voltage determination circuit **510** is coupled to the voltage regulator **520**. The voltage regulator **520** includes a pulse width modulator (PWM) **522**, a high side (HS) transistor, a low side (LS) transistor, an inductor **L1**, and a capacitor **C1**. The HS transistor couples to the LS transistor at a switch node **525**. One terminal of the inductor **L1** couples to the switch node **525**, and the capacitor **C1** couples to the other terminal of the inductor **L1**. The connection point between the inductor **L1** and the capacitor **C1** is the output of the voltage regulator **520** and provides the bias voltage **130**. The voltage regulator **520** in FIG. 5 is implemented as a switching regulator and specifically a buck converter. However, the voltage regulator **520** can be implemented as other types of switching regulators, linear (non-switching) regulators, or any suitable type of circuit that produces a certain bias voltage **130** responsive to the bias control signal **220** received by the bias voltage determination circuit **510**. In one example, bias voltage generator **120** is implemented as a digital-to-analog converter (DAC), which generates an analog output voltage based on the digital bias control signal **220**. The DAC's analog output voltage would be the bias voltage **130**.

[0030] The PWM **522** reciprocally toggles on and off the HS and LS transistors to produce a square wave on the switch node **525**, and through the inductor **L1** and capacitor **C1** results in a regulated bias voltage **130**. The magnitude of the bias voltage **130** is a function of the magnitude of **VCC** and the duty cycle implemented by the PWM's control of the HS and LS transistors. Thus, by varying the duty cycle implemented by the PWM **522**, the magnitude of the bias voltage **130** can be varied. The bias voltage determination circuit **510** generates a control signal **515** to the PWM **522**. Control signal **515** informs the PWM **522** about the duty cycle to be implemented by the PWM **522** and thus the magnitude of the bias voltage **130**. In other implementations, the control signal **515** may specify the on/off time for either or both of the HS and LS transistors.

[0031] FIG. 6 shows an example of a PLM **610** comprising multiple PLM elements **615** arranged in, for example, an array. A digital controller **605** is coupled to the PLM **610**. Each PLM element **615** may comprise the MEMS structure **100** described above. An array of such MEMS structures **100** may be implemented on a semiconductor die comprising the PLM **610**. The voltage on the electrode **110** of a given PLM element **615** may be the same or different as the voltage on another PLM element **615**. The PLM elements **615** on the PLM **610** are individually addressable. In one example, three primary colors (e.g., red, green, blue) are sequentially shined (e.g., directed or projected) on the mirrors **102** of the PLM elements **615** at a rate fast enough that the human eye integrates the sequence of colors into one target color. FIG. 6 shows three light sources **651**, **652**, and **653**. In one example, the light source sources comprise laser diodes (e.g., light source **651** is a red laser diode, light source **652** is a green laser diode, and light source **653** is a blue laser diode). The light sources **651-653** may be sequenced with only one of the light sources active at any point in time. Other examples may include more than three light sources, for example, multiple laser diodes within the wavelength band generally corresponding to red, multiple laser diodes within the wavelength band generally corresponding to green, and multiple laser diodes within the wavelength band generally corresponding to blue. In another example, multiple PLMs **610** may be provided—one per the three primary colors.

[0032] The digital controller **605** provides image data **606** to the PLM **610**. The image data **606** comprises values that indicate the voltage to generated by the electrode **110** of each PLM element **615**. In the example of FIG. 6, the PLM **610** also implements wavelength control by biasing the conductive plate **106** of each PLM element **615** can be biased to a certain voltage based on the wavelength of light provided to the PLM elements **615**. The digital controller **605** provides the bias

control signal **220** to the bias voltage generator **120**. The bias control signal **220** in this example includes a value indicative of the color of light shined on the PLM **610**. The three primary colors have different wavelengths and thus different half wavelengths. By adjusting the bias voltage differently for the three wavelengths, the displacement between the electrode **110** and the conductive plate **106** can be tuned independently for each color wavelength.

[0033] The PLM **610** also includes a bias voltage generator **120** implemented, for example, in accordance with the hinge plate bias voltage generator **120** shown in FIG. 5. In one example, the bias voltage determination circuit **510** of FIG. 5 is implemented as a hardware look-up table (LUT) in FIG. 6 (or any suitable type of storage element), examples of which are shown in FIGS. 7 and 8. The LUT **700** of FIG. 7 includes a value **705** indicative of a target bias voltage for each color (e.g., 0.1V for red, 0.2V for green, and 0.3V for blue). The bias control signal **220** may include a two-bit binary value **704** which indicates which of the three primary colors is shined onto the PLM **610** at any point in time. In the example of FIG. 7, '00' designates red, '01' designates green, and '10' designates blue. The bias voltage determination circuit **510** accesses the LUT **700** to determine which bias voltage **130** to provide to the conductive plates **160** of the PLM elements **615**. Based on the value **705** from the LUT **700**, the bias voltage determination circuit **510** generates a corresponding control signal **515** to the voltage regulator **520** to thereby cause the voltage regulator to generate the desired bias voltage **130**.

[0034] FIG. 8 illustrates another example of a LUT **800** within the bias voltage determination circuit **510**. In this example, a value **805** indicates the bias voltage for one of the colors such as red, and also specifies a delta voltage **810** through which the bias voltage for the other two colors can be calculated. The delta voltage **810** represents a change in bias voltage for a certain difference in wavelength. In the example of FIG. 8, the delta voltage is 0.1V and represents a change in bias voltage relative to the bias voltage for red of 0.1V for each (for example) 100 nanometers of change in wavelength. The bias voltage for green can be calculated as the bias voltage for red (0.1V in this example) plus the product 0.1V and the difference between the wavelength of green relative the wavelength of red. The bias voltage generator **120** implements the logic to use LUT **800** to determine the bias voltage for a given wavelength of light. The bias control signal **220** in this example indicates the wavelength.

[0035] In yet another example, the bias control signal **220** may include a single bit for red, a single bit for green, and a single bit for blue. The digital controller **605** sets the bit to a value, for example, '1' for the particular color being provided at that point in time, with the other two bits cleared to '0.'

[0036] In FIG. 6, the control voltages to the electrode **110** remain unchanged, but the bias voltage to each conductive plate depends on the color being represented at any point in time. Accordingly, the potential difference between the electrode **110** and the conductive plate **106** is the voltage on the electrode **110** minus the bias voltage applied to the conductive plate **106**.

[0037] The displacement between the conductive plate **106** and the electrode **110** for a given electrode **110** voltage may be influenced by temperature of the PLM **610**. That is, at higher temperatures for a given electrode voltage, the conductive plate **106** may fall nearer the electrode **110** thereby reducing the distance between the conductive plate and the electrode. FIG. 9 shows the use of a PLM **910** that implements temperature compensation. FIG. 9 shows the digital controller **605** coupled to PLM **910**. The PLM **910** includes multiple PLM elements **615** as described above, a temperature sensor **920**, a temperature scale circuit **930**, and the bias voltage generator **120**. The temperature sensor may be implemented as, for example, a temperature diode, a thermistor, or a resistor with a suitable large temperature coefficient to provide adequate signal-to-noise ratio. The temperature sensor **920** provides an analog temperature signal (TEMP) **925** to the temperature scale circuit **930**.

[0038] Through testing, the amount of change in the distance between the conductive plate **106** and the electrode **110** due to temperature can be determined. For example, the PLM **910** can be placed

in a thermal chamber and heated to different and controlled temperatures. Any of a variety of devices can be used to measure the initial displacement between electrode **110** and conductive plate **106** (e.g., the displacement with no applied voltage to electrode or conductive plate). Examples of such devices include laser Doppler vibrometers, and interferometers. From the temperature testing, a temperature coefficient for the PLM **910** can be determined in units of volts per degree Celsius. To determine a bias voltage for a particular temperature level, the temperature level would be multiplied by the temperature coefficient. The temperature scale circuit **930** scales and converts the analog temperature signal **925** to the bias control signal **220**, which in this example is a digital signal. The scaling implemented by the temperature scale circuit **930** is a function of the temperature coefficient. In the example shown in FIG. **9**, the temperature scale circuit **930** includes an amplifier **931**, a LUT **932**, and an analog-to-digital converter (ADC) **933**. The amplifier **931** implements a gain programmable by the LUT **932**. The temperature coefficient determined a priori for the PLM **910** is stored in the LUT **932**. The temperature coefficient can be stored in other types of storage elements as well, such any type of read-only memory, electronic fuses, etc.

[0039] The temperature signal **925** is amplified by the amplifier **931** and thus scaled according to the temperature coefficient. The resulting scaled temperature signal **934** is converted to a digital representation by ADC **933** to thereby generate the bias control signal **220**. The bias control signal **220** in this example is indicative of a bias voltage for the bias voltage generator **120** to generate based on a temperature reading from the temperature sensor **925**.

[0040] In the example of FIG. **9**, both the temperature sensor **920** and the temperature scale circuit **930** are provided on the same semiconductor die as the PLM elements **615**. However, either or both of the temperature sensor **920** and the temperature scale circuit **930** can be implemented external to PLM **910** (i.e., on a separate die). In one embodiment, the temperature sensor **920** is implemented on the PLM's die, and the temperature scale circuit **930** is implemented on a separate die. In another embodiment, the temperature scale circuit **930** is implemented on the PLM's die, and the temperature sensor **920** is implemented on a separate die. In yet another embodiment, both the temperature sensor **920** and the temperature scale circuit **930** are implemented on a different die than the PLM **910** (either the temperature sensor **920** and the temperature scale circuit **930** are implemented on the same die or on separate dies).

[0041] Speckling is a visual artifact that may result from a relatively narrow bandwidth of the laser used as the light source for a PLM. Speckling may result because of the use of a coherent light source (narrow bandwidth light source) with the light reflecting off of mirror **102** impinging on a surface that is not uniformly flat (e.g., has small perturbations across its surface). Speckling manifests itself as an image that appears grainy. FIG. **10** shows an example of a PLM **1010** that implements despeckling. A despeckler control signal **1011** may be an externally supplied control signal to the PLM **1010**. The despeckler control signal **1011** commands the PLM **1010** to turn its despeckling capability on or off. The PLM **1010** in this example includes a time varying bias voltage synthesis circuit **1020** which generates the bias control signal **220**. When despeckling is turned on (enabled), the time varying bias voltage synthesis circuit **1020** generates time varying bias control signal **220** to cause the bias voltage generator **120** to thereby generate time varying bias voltages **130**. By varying (e.g., at a rate fast enough not to be perceptible by human vision) the bias voltage to the conductive plates **106** of the PLM elements **615**, speckling is reduced or avoided altogether.

[0042] In one embodiment, the time varying bias voltage synthesis circuit **1020** is preprogrammed to output a sequence of bias control signals **220** that correspond to different voltages. In one example, the time varying bias voltage synthesis circuit **1020** includes a storage element (e.g., any suitable type of read only memory, LUT, etc.) that stores a sequence of values. The values are sequentially read from the storage element and provided to the bias voltage generator as a sequence of bias control signals **220**.

[0043] FIG. **11**, for example, illustrates the sequence of values from the storage element of the time

varying bias voltage synthesis circuit **1020** implementing a stair-step waveform that approximates a triangle wave. The bias voltage regulator **120** generates a bias voltage **130** corresponding to each time varying bias voltage synthesis circuit **1020** step of the triangle wave. In one such implementation, one complete cycle of the triangle wave values are stored in the storage element, and the sequence is repeated over and over to cause the bias voltage generator **120** to generate a sequence of correspondingly sized bias voltages **130**. A clock operates the time varying bias voltage synthesis circuit **1020** to output the values from the storage element at a particular rate.

[0044] FIG. **12** illustrates a sequence of values from the storage element of the time varying bias voltage synthesis circuit **1020** that synthesizes a sinusoidal signal. The shape of the waveform can be any suitable time varying signal (triangle wave, sine wave, square wave, etc.).

[0045] At least some spatial light modulators operate on the basis of constructive and destructive interference and diffraction effects and thus depend upon using a coherent illumination source (e.g., laser). Therefore, if the wavelength of the illumination source changes, which can happen due to temperature, aging, or mode hopping, the ability of the spatial light modulator to display the desired content may be impaired. If the wavelength shifts lower, then bias voltage should be decreased, and if the wavelength shifts higher, then bias voltage should be increased.

[0046] FIG. **13** shows an embodiment in which a PLM **1310** includes a wavelength sensor **1320** coupled to an amplifier **1330**. The wavelength sensor **1320** may be any suitable sensor that produces an electrical signal responsive to the wavelength of light received on the sensor. In one example, the wavelengths sensor **1320** is a spectroradiometer. The amplifier **1330** receives the output signal **1321** from the wavelength sensor **1320** on one input, and a reference signal (VREF) on another input. The amplifier **1330** amplifies the difference between signal **1321** from the wavelength sensor **1320** and VREF. The output signal **1335** is a signal that indicates whether the signal **1321** is higher or lower than VREF and the magnitude of the difference. Responsive to the signal **1321**, the bias voltage generator **120** produces the bias voltage **130** to account for changes in the wavelength of a light source (e.g., due to age of the light source).

[0047] In another implementation, the change in wavelength of the light source may be due to temperature changes of the light source. Accordingly, a temperature sensor may be coupled to or near the light source, and the temperature signal used by the bias voltage generator **120** to generate the bias voltage **130**.

[0048] If a phase light modulator is subject to mechanical vibration, the phase light modulator may undesirably cause a conjugate ghost image to be displayed. A conjugate ghost image is a copy of the desired image spatially shifted from the desired image and inverted. Vibration causes a change in the displacement of the conductive plate **106** relative the respective electrode **110**, and thus the displacement of each mirror is not what is otherwise intended. As a result of unintended mirror displacements, conjugate images can be created. FIG. **14** shows an example of a PLM **1410** that includes a vibration sensor **1420** coupled to a vibration feedback circuit **1430**. The vibration sensor **1420** may be an accelerometer or other type of sensor that outputs a signal responsive to mechanical vibration. The analog signal **1425** from the vibration sensor **1420** is provided to the vibration feedback circuit **1430**. A vibration control signal **1431** enables or disables the vibration control feedback of the bias voltage **130**. If vibration control is enabled (e.g., via the vibration control signal **1431**), the vibration feedback circuit **1430** generates values for the bias control signal **220** that cause the bias voltage generator **120** to generate bias voltages to counteract the vibration to thereby reduce or eliminate any conjugate ghost images.

[0049] A voltage signal in phase with the vibration experienced by the PLM may be inverted and provided back into the PLM. In some implementations, the base resonance of the PLM may be higher than that of the mechanical vibrations sensed by the accelerometer. Accordingly, the vibration feedback circuit **1430** may implement an application-specific transfer function based on the anticipated mechanical oscillation expected by the PLM at a certain frequency.

[0050] As described above, the phase light modulator includes a conductive plate **106** that moves

vertically with respect to the electrode **110** responsive to an applied potential difference. The separation distance between the conductive plate **106** and the electrode **110** (identified as D2 in FIG. **3**) is nominally a function of the potential difference. Referring briefly to FIG. **1**, the height of the support post **108** is D1. Due to manufacturing tolerances, the height of the support posts **108** can vary from device to device. For example, D1 may be 5 nm with a tolerance of ± 1 nm. That is, D1 may be in the range of 4 nm to 6 nm across a sampling of devices. The separation D2 thus is a function of both (a) the potential difference between the conductive plate **106** and the electrode **110** and (b) the height D1 of the support posts **108**. Accordingly, unfortunately the separate D2 for a given potential difference can vary from device to device according to the manufacturing tolerance of the height of the support posts **108**.

[0051] Electrostatic actuators have a “pull-in” voltage, which is the potential difference between electrode **110** and conductive plate **106** to cause the mirror **102** to be separated by a distance from the electrode **110**, which is equal to one-half the wavelength of the respective wavelength received reflected by the mirror.

[0052] The pull-in point (one-half wavelength separation between mirror **102** and electrode **110**) is a function, in part, of the height D1 of the support posts **108**. Accordingly, for different height support posts **108**, the pull-in voltage will be different. FIG. **15** shows an example in which the PLM **1510** applies a bias voltage to the conductive plates **106** of the PLM elements **615** to calibrate for tolerances of the support posts **108**. In this example, the PLM **1510** includes a storage device **1520** in which binning information is stored. FIG. **16** shows an example of binning information in the form of a LUT **1525**. For each of different pull-in voltages **1610**, the binning information includes a particular bias voltage **1620**. In this example, the pull-in voltage for a given PLM **1510** is used as a proxy value for the height D1 of the support posts **108**. For a larger D1, the pull-in voltage will be higher, and for a smaller D1, the pull-in voltage will be lower. Through testing, a bias voltage can be determined for each of the various pull-in voltages within the binning information. The binning information is stored in storage device **1520**. For a specific PLM device, the pull-in voltage can be measured, for example, using a laser Doppler or vibrometer.

[0053] The light source(s) for a PLM have a bandwidth that is relatively narrow, relatively wide, or somewhere in between. A laser diode, for example, has a relatively narrow bandwidth. Other light sources such as light emitting diodes, have bandwidths that are wider than lasers. In general, a PLM is tuned to a particular wavelength of light. FIG. **17A** illustrates an example of the wavelengths **1701** produced by a light source for a PLM that is tuned to one particular wavelength **1705**. If the light source includes that particular wavelength and other wavelengths due its wider bandwidth, a visual smearing effect may be noticed from using a wider bandwidth light source.

[0054] FIG. **17B** is directed to an embodiment of a PLM **1720** in which the bias voltage for the support posts is modulated across approximately the full bandwidth of the light source so that the PLM is tuned to the various wavelengths of the light source. The PLM **1720** includes a time varying bias voltage synthesis circuit **1730** (implemented the same or similar to the time varying bias voltage synthesis circuit **1020** of FIG. **12**). The time varying bias voltage synthesis circuit **1720** produces the bias control signal **220** to the bias voltage generator **120** such that the bias voltage generator **120** produces a time varying bias voltage to the PLM elements **615** to tune the PLM to various wavelengths within some or all of the light source's bandwidth. In one example, the time varying bias voltage synthesis circuit **1730** causes the bias voltage generator **120** to sweep its bias voltage output at a rate that is faster than the human eye could detect (e.g., 100 to 360 Hz)

[0055] Water vapor may exist in or around the PLM elements. The dipole moment of water on the electrode surface may result in the formation of charge on the surface of the electrode (E1, E2, E3). FIGS. **18A-18C** are directed to an embodiment to reduce or eliminate the charge that may otherwise form from water on the surface of an electrode. In this embodiment, the water-based charge is mitigated by reversing the polarity of the potential difference between the electrode **110** and the conductive plate **106**. FIG. **18A** illustrates the state of the potential difference with +10V

applied to E1-E3 and a bias voltage of 0V applied to the support posts and conductive plate **106**. FIG. **18B** illustrates the reversal of the potential difference with 0V applied to E1-E3 and a bias voltage of 10V applied to the support posts and conductive plate **106**. In FIG. **18A**, the potential difference from the electrodes E1-E3 to the conductive plate **106** is +10V, and in FIG. **18B**, the potential difference from the electrodes E1-E3 to the conductive plate **106** is -10V. By reversing the potential difference, the dipole moment of water on the electrode surface is varied (e.g., pseudo randomized) and charge is reduced or eliminated on the surface of the electrode due to water. [0056] FIG. **18C** shows an example implementation in which each electrode bit (E1 is shown in FIG. **18C**) is provided to an input of an exclusive-OR (XOR) gate **1810**. Another input of XOR gate **1810** receives either a logic 0 or a logic 1 from switch SW1 based on a control signal **1812** from a polarity reversal circuit **1820**. The logic state of E1 is flipped (0 becomes 1, and vice versa) upon XORing E1 with a logic 1. Otherwise, XOR'ing E1 with a 0 maintains the logic state of E1 unchanged. The output signal **1815** from XOR gate **1810** is used as control signal to switch SW3, which provides either 0V or the electrode voltage (e.g., +10V) to the E1 electrode of each PLM element **615**. In one implementation, E1 being a 1 may be intended to cause +10V to be applied to the electrode, whereas E1 being a 0 may be intended to cause 0V to be applied to the electrode. By flipping the logic state of E1, E1 being a 1 may instead cause 0V to be applied to the electrode, and E1 being a 0 may cause +10V to be applied to the electrode. SW2 is used to also cause a polarity reversal for the bias voltage. A control signal **1813** (which may be control signal **1812** or a separate control signal) from the polarity reversal circuit **1820** causes the bias voltage to the support posts **108** to change polarity.

[0057] Accordingly, the polarity reversal circuit **1820** generates control signals **1812** and **1813** to flip the polarity of potential difference between the electrode **110** and the conductive plate **106**. The polarity reversal circuit **1820** may implement a duty cycle for the control signals **1812** and **1813** and thus a duty cycle for polarity reversal of the potential difference between the electrode **110** and the conductive plate **106**. In one example, the duty cycle is 50%, but can be other than 50% in other examples.

[0058] If the spacing between the mirror **102** and the electrode **110** is not precisely controlled, a conjugate ghost image may exist in the resulting image created by the PLM. Conjugate ghost images are described above. FIG. **19** shows an example of an image **1905** containing an intended image **1910** and a conjugate ghost image **1920**. Image **1910** should be displayed, but not conjugate ghost image **1920**. The example of FIG. **19** includes an image processing system **1930** which processes the image **1905** to determine whether a conjugate ghost image exists. In one example, the image processing system **1930** analyzes a portion of image **1905** where an intended image should not exist. This analysis can be performed by comparing the light magnitude of the image detected in the portion of the image **1905** with a threshold value (below the threshold value, no image (conjugate ghost or otherwise) is determined to exist, and above the threshold, a conjugate ghost image is determined to exist by the image processing system **1930**). Responsive to the image processing system **1930** determining that a conjugate ghost image exists, the image processing system **1930** generates a bias control signal **220** to the PLM elements **615** to change the bias voltage to the conductive plate **106**. The image processing system **1930** continues analyzing image **1905** after causing the bias voltage to change. If a conjugate ghost image is still detected, the image processing changes the value of the bias control signal **220** to again change the bias voltage. This process repeats until the image processing system **1930** no longer detects a conjugate ghost image. The image processing system **1930** may be separate from the PLM or may be integrated on the PLM (e.g., an image processing circuit on the PLM with an input that can be connected to an external camera).

[0059] Any of the aforementioned examples for feedback control of the bias voltage **130** for the conductive plate **106** of each PLM element **615** can be combined with any other example. That is, two or more of the aforementioned examples can be implemented in concert on the same PLM.

FIG. 20 shows an example of a bias voltage generator **2010** usable for a PLM in which a bias voltage **130** is generated based on color data, temperature, and binning information. The bias voltage generator **2010** includes a bias voltage determination circuit **2020** and a voltage regulator **520**. Voltage regulator **520** may be implemented as described above.

[0060] The illustrative bias voltage determination circuit **200** includes a LUT **2021**, temperature scale circuit **930**, and storage device **1520** which contains the binning information. The outputs of LUT **2021**, temperature scale circuit **930**, and storage device **1520** are coupled to inputs of an adder **2025**, the output of which is coupled to the voltage regulator **520**. The color data is a value indicative of which color is being shined on the PLM elements **615**. Examples of various ways to represent color are described above. The color data is used as a look-up into the LUT **2021** to provide a value **2031** indicative of the bias voltage corresponding to the color. The value **2031** is provided to the adder **2025**. The temperature signal (TEMP) **925** from temperature sensor **920** is provided to the temperature scale circuit **930** which produces a temperature-based control signal **2032** (equivalent to the bias control signal **220** of FIG. 9) to the adder **2025**. Binning information also is used to provide a control signal **2033** (equivalent to the bias control signal **220** of FIG. 15) to the adder **2025**.

[0061] The adder **2025** adds together the various input signals **2031-2032** to produce a signal **2040** to the voltage regulator **520**. As described above, the signal **2040** may control the duty cycle implemented by the PWM controller of the voltage regulator. The voltage regulator **520** produces a regulated bias voltage **130** based on the signal **2040** from the bias voltage determination circuit **2020**, and the signal **2040** is a function of, in this example, color, temperature, and binning information.

[0062] Although color, temperature, and binning information are combined to produce a bias voltage **130** for the conductive plates **106** of the PLM elements **615**, any two or more of the aforementioned factors can be combined together in a similar manner in other implementations.

[0063] The term “couple” is used throughout the specification. The term may cover connections, communications, or signal paths that enable a functional relationship consistent with the description of the present disclosure. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is directly coupled to device B; or (b) in a second example, device A is indirectly coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0064] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. An apparatus comprising: a phase light modulator; a bias voltage circuit configurable to receive an input signal indicating whether to perform despeckling and to produce a control signal responsive to the input signal; and a bias voltage generator coupled to the bias voltage circuit and to the phase light modulator, the bias voltage generator configurable to produce a bias voltage for the phase light modulator responsive to the control signal.
2. The apparatus of claim 1, wherein the bias voltage circuit is configurable to produce a time varying bias control signal.
3. The apparatus of claim 2, wherein the bias voltage circuit comprises a storage element configurable to store a sequence of values.
4. The apparatus of claim 2, wherein the time varying bias control signal is a sinusoidal signal or a triangular signal.
5. The apparatus of claim 1, further comprising a controller coupled to the phase light modulator,

the controller configurable to produce an electrode voltage signal for the phase light modulator.

6. The apparatus of claim 1, wherein the phase light modulator comprises a mirror, wherein the mirror is configurable to receive the bias voltage.

7. The apparatus of claim 1, further comprising a storage element, wherein the bias voltage circuit is configurable to produce a time varying bias control signal by sequentially reading a sequence of values from the storage element, and wherein the bias voltage generator is further configurable to produce the bias voltage responsive to the time varying bias control signal.

8. The apparatus of claim 1, wherein the phase light modulator comprises: a first electrode configurable to receive the bias voltage; a second electrode configurable to receive a data signal; and a mirror over the first electrode and the second electrode, the mirror coupled to the first electrode.

9. A circuit comprising: a first bias voltage circuit configurable to produce a first bias voltage signal; a second bias voltage circuit configurable to produce a second bias voltage signal; a combining circuit coupled to the first bias voltage circuit and to the second bias voltage circuit, the combining circuit configurable to combine the first bias voltage signal and the second bias voltage signal to produce a third bias voltage signal; and a voltage regulator coupled to the combining circuit, the voltage regulator configurable to produce a bias voltage responsive to the third bias voltage signal.

10. The circuit of claim 9, wherein the combining circuit is an adder.

11. The circuit of claim 9, wherein the first bias voltage circuit comprises a look-up-table.

12. The circuit of claim 9, wherein the first bias voltage circuit is configurable to: receive input data; and produce the first bias voltage signal responsive to the input data.

13. The circuit of claim 12, wherein the input data is color data.

14. The circuit of claim 12, wherein the input data is temperature data.

15. The circuit of claim 9, wherein the first bias voltage circuit is configurable to produce the first bias voltage signal to binning information.

16. The circuit of claim 9, further comprising a phase light modulator coupled to the voltage regulator, the phase light modulator configurable to receive the bias voltage from the voltage regulator.

17. A system comprising: a light source configurable to produce light; a controller configurable to produce an image data signal; a storage element; a bias voltage circuit coupled to the storage element, the bias voltage circuit configurable to produce a time varying bias control signal responsive to sequentially reading a sequence of values from the storage element; a bias voltage generator coupled to the controller, the bias voltage generator configurable to: receive the time varying bias control signal; and produce a time varying bias voltage responsive to the time varying bias control signal; and a phase light modulator coupled to the controller and to the bias voltage generator, the phase light modulator optically coupled to the light source, the phase light modulator configurable to modulate the light responsive to the time varying bias voltage and the image data signal.

18. The system of claim 17, wherein the bias voltage circuit is further configurable to: receive a despeckler control signal; and produce the time varying bias control signal responsive to the despeckler control signal.

19. The system of claim 17, wherein the time varying bias voltage is a sinusoidal signal or a triangular signal.

20. The system of claim 17, wherein the phase light modulator comprises: a first electrode configurable to receive the time varying bias voltage; a second electrode configurable to receive the image data signal; and a mirror over the first electrode and the second electrode, the mirror coupled to the first electrode.
